Rank-27 Linear Import-Order Reconstruction

Tactix engineering handoff

24 August 2026

Executive outcome

Current/current Rank-27 profiling localized a portable common-state throughput loss in Kernel representation: after variable compaction, Tactix repeatedly comparison-sorted every live variable to reconstruct the permanent DIMACS first-occurrence order. At the exact conflict-9450 boundary this helper alone accounted for 3.2251% of production cycles. The general repair records the inverse first-occurrence permutation once, retains retired external identities as holes, and resolves surviving entries through the existing external-to-internal compaction map. Direct Kernel clients keep the previous fallback order. No proof rule, strategy decision, scheduling cadence, effort budget, or benchmark-specific arm changes.

The repaired profile preserves every non-time counter and removes 411 ms total and 402 ms inprocessing by conflict 9450. Its event trace is byte-identical to the accepted current-Tactix trace, including exact rule boundaries, candidate/order images, logical ticks and costs, scheduling order, and resulting learning and pruning. The mandatory full frozen Tier B plus five predeclared mechanism-relevant Tier C guard passes 66/66 with independently checked models and proofs, no solve loss, and zero certificate residue. Exact-source sanitizer fuzz passes 10,000 cases with zero failures. The checked immutable Rank-27 witness preserves every non-time counter and identical proof bytes while improving 548.645 to 413.830 internal seconds. This is a 134.815-second same-trajectory recovery, but the 300-second objective remains open.

Frozen authority and localization

The immutable Rank-27 input is `tierD24-26/5911653e4edadec391654c8f06353f9a-oski15a01b06s_opt.cnf.xz`, SHA-256 `5435b1b271b049ed9291ab1b9b297afb33b8d2d3c50ff1b48bfaed4ddf7bb4e9`. Frozen Kissat 4.0.4 proves UNSAT in 278.187 seconds and was not rerun. The accepted current-Kissat diagnostic is 286.30 seconds, with 188.64 seconds in search, 26.19 in Vivify, 16.32 in elimination, 10.42 in Substitute, and 5.91 in Congruence. The preceding checked current Tactix is valid UNSAT in 548.645 internal / 549.172 wall seconds, with 335.055 seconds in search and 66.680/36.947/36.659/21.752 seconds in Vivify/BVE/Substitute/Congruence. Those full-run totals are localization leads across different trajectories, not same-state proof of a rule defect.

The accepted common-state chain is narrower. Campaigns `0321`–`0323` establish exact current/current state through conflict 9449 and the second Vivify entry: 5,761,854 decisions, all 4,915 candidates in identical order, equal per-candidate logical work, the same 19 binary additions, exact scheduling, and equal portable pruning. The only extra Tactix deletion is normalization of an inherited duplicate; that Vivify semantic lead remains closed.

Observer-free current-Tactix profile `benchmarks/runs/20260824-1121-tierd-rank27-current-vivify-production-profile/` stops at conflict 9450 in 21.051 seconds total, 10.029 seconds

inprocessing, and 11.022 seconds search/scaffolding. It emits 5,795,248 decisions, 9,450 conflicts, and 96,338,812 propagations. A zero-loss `cycles:u` profile assigns 3.2251% to the comparison sort inside `kernel_vars_in_import_order`. Source inspection shows the sort merely reconstructs an order that input parsing had already assigned permanently; no rule or policy owns that representation work.

General repair and exact-state evidence

Kernel now stores an inverse vector from permanent import rank to external variable, complementing its existing external-rank vector. First import appends one external identity. Reconstruction traverses that stable vector, skips retired holes through `external_to_internal`, then appends any unrecorded live direct-client variables in stable external order. The previous semantics are therefore unchanged before and after compaction, but the recurring $O(n \log n)$ comparison sort becomes a linear image traversal. A focused Kernel regression imports variables in non-numeric order, retires one imported and one unrecorded identity, compacts, and checks the exact surviving order and all Kernel invariants.

Production campaign `benchmarks/runs/20260824-1136-tierd-rank27-import-order-linear-profile/`, job `1645537`, uses a separate read-only Release diagnostic pin and the same 16 GiB / two-CPU contract. All non-time statistics match the predecessor at conflict 9450. Total time is 20.640 seconds, inprocessing 9.627, and search/scaffolding 11.012: reductions of 411, 402, and 10 ms respectively. The comparison-sort symbol disappears from 4,209 sampled cycles with zero lost samples. The next sampled owners are propagation 28.87%, backtrack 4.93%, garbage collection 4.51%, heap bubbling 4.07%, vector growth 3.48%, and Congruence gate extraction 3.30%; none is yet an authorized behavioral repair.

Exact event-trace campaign `benchmarks/runs/20260824-1142-tierd-rank27-import-order-linear-trace/`, job `1645547`, emits 64 untruncated rows. They are byte-identical to the accepted current-Tactix predecessor, SHA-256 `7a6216bb86c236512dea03de76c4c9ab2ffdd745bd7ca7b9fa5507e708987cc1`. Thus every observed rule boundary, formula state, candidate identity and order, normalized tick/cost image, scheduling event, and pruning result remains exact through the conflict-9449 Probe and conflict-9450 stop. This explicitly preserves the closed Sweep parity, counted directed bridge repair, BVE antecedent admission, and the proof-rule/strategy boundary.

Mandatory guard and correctness gates

The exact candidate Release binary has SHA-256 `4247d489270604692354bbb630f8590eb9e05168f0e9086a752b4de88f9ae57c`. Mandatory campaign `benchmarks/runs/20260824-1145-BC-import-order-linear-guard/`, array `1645550` and aggregate `1645557`, runs all 28 frozen Tier-B inputs plus the fixed five Tier-C mechanism guards under 16 GiB, 1,200 seconds, and two CPUs. Both candidate and shipped arms validate 17 SAT models and 16 UNSAT proofs: 66/66 results, zero invalid, missing, regression, cleanup, or residue entries. Candidate PAR-2 is 1,734.944 seconds versus 2,443.861 shipped, ratio 0.709919263. Summary SHA-256 is `925e4b3074b73bbefb9808a19ae5854ada3dc5cf3fa833c32ff17829407bf6eb`.

The cumulative Release suite is 73/74. Every executable, proof/model, configuration, trace, static rule/policy boundary, and campaign cleanup test passes; only `doc_size` fails on the already recorded living-document debt. Focused ASan/UBSan `test_kernel` passes. Repository authority makes documentation compaction opt-in, so no evidence is silently removed to manufacture a green count.

Exact-source native-LRAT sanitizer campaign `benchmarks/runs/20260824-1150-import-order-linear-fuzz/`, job `1645587`, completes 10,000 cases with ASan and UBSan while stressing BVE, Substitute, and Sweep across retirement. It has zero failures: 5,044 checked SAT models, 3,381 checked native-LRAT UNSAT proofs, and 1,575 declared five-second nonterminations. The three target rules fire in 5,823, 4,804, and 2,471 cases respectively, totaling 1,329,770 stress-target applications. Twenty-five bounded-timeout CNFs are retained as diagnostic inputs; no proof, LRAT, checker, or certificate residue remains. Random generation did not reach VariableCompaction, so the exact ASan/UBSan Kernel regression remains the direct compaction-with-holes authority. No soundness, rejected proof/model, sanitizer, crash, or invariant trigger fired; the Claude workflow was not invoked.

Tier-D cancellation and checked Rank 27

At user direction, full current Tier-D array/aggregate `1645240`/`1645241` and surviving predecessor `1642950`/`1642951` were canceled to release capacity for fuzzing and convergence. No frozen reference solver was rerun. Current campaign `0343` preserves 294 classified rows: 103 checked SAT models, 84 checked UNSAT proofs, 98 solver timeouts, three `bad_alloc` resource exits, and six CakeML heap-exhaustion checker exits after `drat-trim` verification. The predecessor preserves 1,165 classified rows. Cancellation and resource/checker exits are not solver correctness failures; both partial campaigns are non-scorable. Published JSON and logs remain, while exact result-less canceled scratch was deleted under zero retention. Pending, retained, scratch, and certificate residue are zero in both campaigns; each has a frozen `cancellation.json` disposition.

The checked immutable Rank-27 campaign `benchmarks/runs/20260824-1155-tierd-rank27-import-order-linear-witness/` uses the candidate SHA above, immutable input hash, two CPUs, 16 GiB, a 1,200-second solver limit, pinned `drat-trim`, and zero retention. Job `1645699` returns UNSAT in 413.830 internal / 414.375 wall seconds versus 548.645 / 549.172 for the preceding checked pin: 134.815 internal and 134.797 wall seconds recovered. All 3,813,669 conflicts, 175,272,633 decisions, 2,512,356,477 propagations, rule yields, and every other non-time statistic are exact. The 1,514,395,111-byte proof is also byte-identical, SHA-256 `0f99f55ac55571d6f83fe53ec8cb4a6a4c8df0a6e127442e50f47e25bea8bb2d`. Pinned `drat-trim` verifies it in 112.340 seconds with 20,047,408 resolution steps and zero RAT lemmas in the core. Proof, decompressed input, checker derivatives, and scratch are deleted before result publication.

The savings span every repeated reconstruction after compaction: inprocessing falls 213.590 to 167.753 seconds and search/scaffolding 335.055 to 246.078, while Vivify/BVE/Substitute/Congruence fall to 55.877/31.122/21.951/18.072 seconds. This validates the persistent linear representation mechanism far beyond the early 411-ms profile prefix without attributing changed search: the trajectory and proof are exact. At 413.830 seconds, however, Rank 27 still misses the 300-second target and frozen Kissat's 278.187 seconds.

The repair is a validated general representation improvement, not an argument to reopen completed semantic investigations. If the checked witness remains above 300 seconds, the next worse-first step is to compare the leading propagation/backtrack/garbage/heap owners against current Kissat at an exact common state, preserving exact tick/cost, candidate order, scheduling order, and resulting pruning before changing behavior.